Model for "ch4"

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General Condition

- Basis type: lgs
- SAMB selection:
  - Type: [Q, G]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A_1, A_2, E, T_1, T_2]
  - Spin (s): [0, 1]
- Atomic selection:
  - Type: [Q, G, M, T]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A_1, A_2, E, T_1, T_2]
  - Spin (s): [0, 1]
- Site-cluster selection:
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A_1, A_2, E, T_1, T_2]
- Bond-cluster selection:
  - Type: [Q, G, M, T]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A_1, A_2, E, T_1, T_2]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: false

Group and Unit Cell

- Group: PG No. 31 T_{d} $\bar{4}3m$ [cubic]
- Unit cell:
  $a =$ 1.00000, $b =$ 1.00000, $c =$ 1.00000, $\alpha =$ 90.0, $\beta =$ 90.0, $\gamma =$ 90.0
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ [1.00000, 0.00000, 0.00000]
  $\boldsymbol{a}_2 =$ [0.00000, 1.00000, 0.00000]
  $\boldsymbol{a}_3 =$ [0.00000, 0.00000, 1.00000]

Symmetry Operation

Table 1: Symmetry operation

	#	SO	#	SO	#	SO	#	SO	#	SO
	1	1	2	2_{001}	3	2_{010}	4	2_{100}	5	3^{+}_{111}
	6	3^{+}_{-11-1}	7	3^{+}_{1-1-1}	8	3^{+}_{-1-11}	9	3^{-}_{111}	10	3^{-}_{1-1-1}
	11	3^{-}_{-1-11}	12	3^{-}_{-11-1}	13	m_{1-10}	14	m_{110}	15	-4^{+}_{001}
	16	-4^{-}_{001}	17	m_{01-1}	18	-4^{+}_{100}	19	-4^{-}_{100}	20	m_{011}
	21	m_{-101}	22	-4^{-}_{010}	23	m_{101}	24	-4^{+}_{010}		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_1)$	A_1	0	Q, T	-	-	1
2	$\mathbb{G}_0(A_2)$	A_2	0	G, M	-	-	1
3	$\mathbb{G}_3(A_2)$	A_2	3	G, M	-	-	$\sqrt{15}xyz$
4	$\mathbb{G}_{2,1}(E)$	E	2	G, M	-	1	$\frac{\sqrt{3}(-x^2+y^2)}{2}$
5	$\mathbb{G}_{2,2}(E)$					2	$\frac{-x^2-y^2+2z^2}{2}$
6	$\mathbb{Q}_{2,1}(E)$	E	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
7	$\mathbb{Q}_{2,2}(E)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
8	$\mathbb{G}_{1,1}(T_1)$	T_1	1	G, M	-	1	x
9	$\mathbb{G}_{1,2}(T_1)$					2	y
10	$\mathbb{G}_{1,3}(T_1)$					3	z
11	$\mathbb{G}_{2,1}(T_1)$	T_1	2	G, M	-	1	$\sqrt{3}yz$
12	$\mathbb{G}_{2,2}(T_1)$					2	$\sqrt{3}xz$
13	$\mathbb{G}_{2,3}(T_1)$					3	$\sqrt{3}xy$
14	$\mathbb{G}_{3,1}(T_1)$	T_1	3	G, M	-	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
15	$\mathbb{G}_{3,2}(T_1)$					2	$\frac{y\left(-3x^2+2y^2-3z^2\right)}{2}$
16	$\mathbb{G}_{3,3}(T_1)$					3	$\frac{z\left(-3x^2-3y^2+2z^2\right)}{2}$
17	$\mathbb{Q}_{1,1}(T_2)$	T_2	1	Q, T	-	1	x
18	$\mathbb{Q}_{1,2}(T_2)$					2	y
19	$\mathbb{Q}_{1,3}(T_2)$					3	z
20	$\mathbb{Q}_{2,1}(T_2)$	T_2	2	Q, T	-	1	$\sqrt{3}yz$
21	$\mathbb{Q}_{2,2}(T_2)$					2	$\sqrt{3}xz$
22	$\mathbb{Q}_{2,3}(T_2)$					3	$\sqrt{3}xy$
23	$\mathbb{G}_{3,1}(T_2)$	T_2	3	G, M	-	1	$\frac{\sqrt{15}x\left(y^2-z^2\right)}{2}$
24	$\mathbb{G}_{3,2}(T_2)$					2	$\frac{\sqrt{15}y\left(-x^2+z^2\right)}{2}$
25	$\mathbb{G}_{3,3}(T_2)$					3	$\frac{\sqrt{15}z\left(x^2-y^2\right)}{2}$

Table 3: dimension = 16

	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
	0	$\|s,\uparrow\rangle$ @C(1)	1	$\|s,\downarrow\rangle$ @C(1)	2	$\|p_x,\uparrow\rangle$ @C(1)	3	$\|p_x,\downarrow\rangle$ @C(1)	4	$\|p_y,\uparrow\rangle$ @C(1)
	5	$\|p_y,\downarrow\rangle$ @C(1)	6	$\|p_z,\uparrow\rangle$ @C(1)	7	$\|p_z,\downarrow\rangle$ @C(1)	8	$\|s,\uparrow\rangle$ @H(1)	9	$\|s,\downarrow\rangle$ @H(1)
	10	$\|s,\uparrow\rangle$ @H(2)	11	$\|s,\downarrow\rangle$ @H(2)	12	$\|s,\uparrow\rangle$ @H(3)	13	$\|s,\downarrow\rangle$ @H(3)	14	$\|s,\uparrow\rangle$ @H(4)
	15	$\|s,\downarrow\rangle$ @H(4)								

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|s\rangle$	1
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z

SAMB: 120 (all 120)

- `C` : 'C' site-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `1o`

  z1 $\quad \mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

- `C` : 'C' site-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `1o`

z10 $\mathbb{G}_{0}^{(1,1;c)}(A_2) = \mathbb{G}_{0}^{(1,1;a)}(A_2)\mathbb{Q}_{0}^{(s)}(A_1)$

z15 $\mathbb{G}_{2,1}^{(1,-1;c)}(E) = \dfrac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

z16 $\mathbb{G}_{2,2}^{(1,-1;c)}(E) = \dfrac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

z37 $\mathbb{G}_{2,1}^{(1,-1;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z38 $\mathbb{G}_{2,2}^{(1,-1;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z39 $\mathbb{G}_{2,3}^{(1,-1;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z73 $\mathbb{Q}_{1,1}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z74 $\mathbb{Q}_{1,2}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z75 $\mathbb{Q}_{1,3}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z76 $\mathbb{Q}_{1,1}^{(1,0;c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z77 $\mathbb{Q}_{1,2}^{(1,0;c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

z78 $\mathbb{Q}_{1,3}^{(1,0;c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

- C : 'C' site-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: 1o

  z2 $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

  z3 $\mathbb{Q}_0^{(1,1;c)}(A_1) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

  z17 $\mathbb{Q}_{2,1}^{(c)}(E) = \dfrac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$

  z18 $\mathbb{Q}_{2,2}^{(c)}(E) = \dfrac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$

  z19 $\mathbb{Q}_{2,1}^{(1,-1;c)}(E) = \dfrac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$

  z20 $\mathbb{Q}_{2,2}^{(1,-1;c)}(E) = \dfrac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$

  z40 $\mathbb{G}_{1,1}^{(1,0;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$

  z41 $\mathbb{G}_{1,2}^{(1,0;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$

  z42 $\mathbb{G}_{1,3}^{(1,0;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$

  z79 $\mathbb{Q}_{2,1}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$

  z80 $\mathbb{Q}_{2,2}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$

  z81 $\mathbb{Q}_{2,3}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$

  z82 $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$

z83 $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$

z84 $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$

- `H` : 'H' site-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `4a`

  z4 $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

  z85 $\mathbb{Q}_{1,1}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(T_2)}{3}$

  z86 $\mathbb{Q}_{1,2}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(T_2)}{3}$

  z87 $\mathbb{Q}_{1,3}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,3}^{(s)}(T_2)}{3}$

- `H;C_001_1` : 'C'-'H' bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `4a@4a`

  z5 $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

  z11 $\mathbb{G}_0^{(1,-1;c)}(A_2) = \dfrac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)\right)}{3}$

  z21 $\mathbb{G}_{2,1}^{(1,-1;c)}(E) = \dfrac{-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)}{2}$

  z22 $\mathbb{G}_{2,2}^{(1,-1;c)}(E) = \dfrac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)\right)}{6}$

  z43 $\mathbb{G}_{1,1}^{(1,-1;c)}(T_1) = \dfrac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_0^{(b)}(A_1)}{3}$

z44 $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{0}^{(b)}(A_1)}{3}$$

z45 $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{0}^{(b)}(A_1)}{3}$$

z46 $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)\right)}{6}$$

z47 $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$$

z48 $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$$

z88 $$\mathbb{Q}_{1,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3}$$

z89 $$\mathbb{Q}_{1,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(T_2)}{3}$$

z90 $$\mathbb{Q}_{1,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3}$$

z91 $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)\right)}{6}$$

z92 $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$$

z93 $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$$

- `H;C_001_1` : 'C'-'H' bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `4a@4a`

z6 $$\mathbb{Q}_{0}^{(c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)\right)}{3}$$

z7 $$\mathbb{Q}_{0}^{(1,0;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)\right)}{3}$$

z12 $$\mathbb{G}_{3}^{(1,-1;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2)\right)}{3}$$

z13 $$\mathbb{G}_{0}^{(1,1;c)}(A_2) = \mathbb{G}_{0}^{(1,1;a)}(A_2)\mathbb{Q}_{0}^{(b)}(A_1)$$

z23 $$\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{3}\left(-\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) + 2\mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)\right)}{6}$$

z24 $$\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{2}$$

z25 $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E) = \frac{-\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2)}{2}$$

z26 $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2) - \mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2) + 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2)\right)}{6}$$

z27 $$\mathbb{Q}_{2,1}^{(1,0;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)\right)}{6}$$

z28 $$\mathbb{Q}_{2,2}^{(1,0;c)}(E) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{2}$$

z29 $$\mathbb{G}_{2,1}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

z30 $$\mathbb{G}_{2,2}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

z49 $$\mathbb{G}_{1,1}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) - \mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{6}$$

z50 $$\mathbb{G}_{1,2}^{(c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z51 $$\mathbb{G}_{1,3}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z52 $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{10}\left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{30}$$

z53 $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{10}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{30}$$

z54 $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{10}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2) + 2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,3}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{30}$$

z55 $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z56 $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z57 $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z58 $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_1) = -\frac{\sqrt{5}\left(3\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) + 2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) + 2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{30}$$

z59 $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{5}\left(3\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) - 2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) - 2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{30}$$

z60 $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{5}\left(-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,3}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{15}$$

z61 $$\mathbb{G}_{1,1}^{(1,0;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) - \mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{6}$$

z62 $$\mathbb{G}_{1,2}^{(1,0;c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z63 $$\mathbb{G}_{1,3}^{(1,0;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z64 $$\mathbb{G}_{1,1}^{(1,1;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3}$$

z65 $$\mathbb{G}_{1,2}^{(1,1;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{3}$$

z66 $$\mathbb{G}_{1,3}^{(1,1;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3}$$

z94 $$\mathbb{Q}_{1,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z95 $$\mathbb{Q}_{1,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z96 $$\mathbb{Q}_{1,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z97 $$\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{6}$$

z98 $$\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z99 $$\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,1}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z100 $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) - \mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{6}$$

z101 $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) - \mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z102 $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(2\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,3}^{(b)}(T_2) - \mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z103 $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z104 $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z105 $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

z106 $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)\right)}{6}$$

z107 $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z108 $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)\right)}{6}$$

z109 $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_2) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) + 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2)}{6}$$

z110 $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_2) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) + 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,3}^{(b)}(T_2) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)}{6}$$

z111 $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_2) = \frac{-\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{1,3}^{(b)}(T_2) - \mathbb{G}_{2,1}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,2}^{(b)}(T_2) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_1)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3}$$

- `H;H_001_1` : 'H'-'H' bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `6a@6b`

z8 $$\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$$

z9 $$\mathbb{Q}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{1,3}^{(b)}(T_1)\right)}{3}$$

z14 $$\mathbb{G}_{0}^{(1,-1;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)\right)}{3}$$

z31 $$\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

z32 $$\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

z33 $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{1,3}^{(b)}(T_1)\right)}{6}$$

z34 $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1)}{2}$$

z35 $$\mathbb{G}_{2,1}^{(1,-1;c)}(E) = \frac{-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)}{2}$$

z36 $$\mathbb{G}_{2,2}^{(1,-1;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)\right)}{6}$$

z67 $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1)\right)}{6}$$

z68 $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1)\right)}{6}$$

z69 $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1)\right)}{6}$$

z70 $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)\right)}{6}$$

z71 $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$$

z72 $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$$

z112 $$\mathbb{Q}_{1,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3}$$

z113 $\mathbb{Q}_{1,2}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(T_2)}{3}$

z114 $\mathbb{Q}_{1,3}^{(c)}(T_2) = \dfrac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3}$

z115 $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)\right)}{6}$

z116 $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$

z117 $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)\right)}{6}$

z118 $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1)\right)}{6}$

z119 $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1)\right)}{6}$

z120 $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{1,1}^{(b)}(T_1)\right)}{6}$

Atomic SAMB

- bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
- ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$

x1 $\mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$

x2 $\mathbb{M}_{1,1}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x3 $\mathbb{M}_{1,2}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x4 $\mathbb{M}_{1,3}^{(1,-1;a)}(T_1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$

- bra: $\langle s,\uparrow|\,,\ \langle s,\downarrow|$
- ket: $|p_x,\uparrow\rangle\,,\ |p_x,\downarrow\rangle\,,\ |p_y,\uparrow\rangle\,,\ |p_y,\downarrow\rangle\,,\ |p_z,\uparrow\rangle\,,\ |p_z,\downarrow\rangle$

x5 $\mathbb{Q}_{1,1}^{(a)}(T_2) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x6 $\mathbb{Q}_{1,2}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

x7 $\mathbb{Q}_{1,3}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

x8 $\mathbb{Q}_{1,1}^{(1,0;a)}(T_2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

x9 $\mathbb{Q}_{1,2}^{(1,0;a)}(T_2) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

x10 $\mathbb{Q}_{1,3}^{(1,0;a)}(T_2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

x11 $\mathbb{G}_{2,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x12 $\mathbb{G}_{2,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$

x13 $\mathbb{G}_{2,1}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

x14 $\mathbb{G}_{2,2}^{(1,-1;a)}(T_1) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

x15 $\mathbb{G}_{2,3}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

x16 $\mathbb{G}_{0}^{(1,1;a)}(A_2) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$

x17 $\mathbb{M}_{2,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

x18 $\mathbb{M}_{2,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

x19 $\mathbb{M}_{2,1}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

x20 $\mathbb{M}_{2,2}^{(1,-1;a)}(T_1) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

x21 $\mathbb{M}_{2,3}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x22 $\mathbb{M}_{0}^{(1,1;a)}(A_2) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$

x23 $\mathbb{T}_{1,1}^{(a)}(T_2) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x24 $\mathbb{T}^{(a)}_{1,2}(T_2) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

x25 $\mathbb{T}^{(a)}_{1,3}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$

x26 $\mathbb{T}^{(1,0;a)}_{1,1}(T_2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

x27 $\mathbb{T}^{(1,0;a)}_{1,2}(T_2) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

x28 $\mathbb{T}^{(1,0;a)}_{1,3}(T_2) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle p_x, \uparrow|$, $\langle p_x, \downarrow|$, $\langle p_y, \uparrow|$, $\langle p_y, \downarrow|$, $\langle p_z, \uparrow|$, $\langle p_z, \downarrow|$
- ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

x29 $\mathbb{Q}^{(a)}_{0}(A_1) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$

x30 $\mathbb{Q}^{(a)}_{2,1}(E) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x31 $\mathbb{Q}^{(a)}_{2,2}(E) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x32 $\mathbb{Q}^{(a)}_{2,1}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

x33 $\mathbb{Q}^{(a)}_{2,2}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x34 $\mathbb{Q}^{(a)}_{2,3}(T_2) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x35 $\mathbb{Q}_{2,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

x36 $\mathbb{Q}_{2,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

x37 $\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

x38 $\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

x39 $\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x40 $\mathbb{Q}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

x41 $\mathbb{G}_{1,1}^{(1,0;a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

x42 $\mathbb{G}_{1,2}^{(1,0;a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

x43 $\mathbb{G}_{1,3}^{(1,0;a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x44 $\mathbb{M}_{1,1}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

x45 $\mathbb{M}_{1,2}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x46 $\mathbb{M}_{1,3}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x47 $\mathbb{M}_3^{(1,-1;a)}(A_2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$

x48 $\mathbb{M}_{1,1}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

x49 $\mathbb{M}_{1,2}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

x50 $\mathbb{M}_{1,3}^{(1,-1;a)}(T_1) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

x51 $\mathbb{M}_{3,1}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{5}}{5} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{5} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$

x52 $\mathbb{M}_{3,2}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{5} & -\frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$

x53 $\mathbb{M}_{3,3}^{(1,-1;a)}(T_1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$

x54 $\mathbb{M}_{3,1}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$

x55 $\mathbb{M}_{3,2}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$

x56 $\mathbb{M}_{3,3}^{(1,-1;a)}(T_2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

x57 $\mathbb{M}_{1,1}^{(1,1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$

x58 $\mathbb{M}_{1,2}^{(1,1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$

x59 $\mathbb{M}_{1,3}^{(1,1;a)}(T_1) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$

x60 $\mathbb{T}_{2,1}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

x61 $\mathbb{T}_{2,2}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$

x62 $\mathbb{T}_{2,1}^{(1,0;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

x63 $\mathbb{T}^{(1,0;a)}_{2,2}(T_2) = \begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

x64 $\mathbb{T}^{(1,0;a)}_{2,3}(T_2) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `1o`

y1 $\mathbb{Q}^{(s)}_0(A_1) = [1]$

** Wyckoff: `4a`

y2 $\mathbb{Q}^{(s)}_0(A_1) = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$

y3 $\mathbb{Q}^{(s)}_{1,1}(T_2) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

y4 $\mathbb{Q}^{(s)}_{1,2}(T_2) = \left[\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$

y5 $\mathbb{Q}^{(s)}_{1,3}(T_2) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right]$

- Bond cluster

** Wyckoff: `4a@4a`

y6 $\mathbb{Q}_{0}^{(b)}(A_1) = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$

y7 $\mathbb{T}_{0}^{(b)}(A_1) = \left[\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, \frac{i}{2}\right]$

y8 $\mathbb{Q}_{1,1}^{(b)}(T_2) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

y9 $\mathbb{Q}_{1,2}^{(b)}(T_2) = \left[\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$

y10 $\mathbb{Q}_{1,3}^{(b)}(T_2) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right]$

y11 $\mathbb{T}_{1,1}^{(b)}(T_2) = \left[\frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}\right]$

y12 $\mathbb{T}_{1,2}^{(b)}(T_2) = \left[\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}\right]$

y13 $\mathbb{T}_{1,3}^{(b)}(T_2) = \left[\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}\right]$

** Wyckoff: `6a@6b`

y14 $\mathbb{Q}_{0}^{(b)}(A_1) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

y15 $\mathbb{Q}_{2,1}^{(b)}(E) = \left[-\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

y16 $\mathbb{Q}_{2,2}^{(b)}(E) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, 0, 0\right]$

y17 $\mathbb{M}_{1,1}^{(b)}(T_1) = \left[0, 0, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, \frac{i}{2}\right]$

y18 $\mathbb{M}^{(b)}_{1,2}(T_1) = \left[-\frac{i}{2}, \frac{i}{2}, 0, 0, \frac{i}{2}, \frac{i}{2}\right]$

y19 $\mathbb{M}^{(b)}_{1,3}(T_1) = \left[\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0\right]$

y20 $\mathbb{Q}^{(b)}_{1,1}(T_2) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, 0, 0, 0\right]$

y21 $\mathbb{Q}^{(b)}_{1,2}(T_2) = \left[0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, 0\right]$

y22 $\mathbb{Q}^{(b)}_{1,3}(T_2) = \left[0, 0, 0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

y23 $\mathbb{T}^{(b)}_{1,1}(T_2) = \left[0, 0, \frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}\right]$

y24 $\mathbb{T}^{(b)}_{1,2}(T_2) = \left[\frac{i}{2}, -\frac{i}{2}, 0, 0, \frac{i}{2}, \frac{i}{2}\right]$

y25 $\mathbb{T}^{(b)}_{1,3}(T_2) = \left[\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, 0, 0\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	C	$\|s,\uparrow\rangle$, $\|s,\downarrow\rangle$, $\|p_x,\uparrow\rangle$, $\|p_x,\downarrow\rangle$, $\|p_y,\uparrow\rangle$, $\|p_y,\downarrow\rangle$, $\|p_z,\uparrow\rangle$, $\|p_z,\downarrow\rangle$
2	H	$\|s,\uparrow\rangle$, $\|s,\downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	C	H	[1]	[s,p]	[s]
2	H	H	[1]	[s]	[s]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'C' (#1) site cluster (1o), -43m

SL	position ($\boldsymbol{s}$)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 8: 'H' (#2) site cluster (4a), .3m

SL	position ($\boldsymbol{s}$)	mapping
1	[0.33333, 0.33333, 0.33333]	[1,5,9,13,17,21]
2	[-0.33333,-0.33333, 0.33333]	[2,7,12,14,19,24]
3	[-0.33333, 0.33333,-0.33333]	[3,8,10,16,18,23]
4	[0.33333,-0.33333,-0.33333]	[4,6,11,15,20,22]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'C'-'H' [1] (#1) bond cluster (4a@4a), D, $|\boldsymbol{v}|$= 0.57735 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.33333,-0.33333,-0.33333]	[0.16667, 0.16667, 0.16667]	[1,5,9,13,17,21]	(1,1)	(1,1)	[0,0,0]
2	[0.33333, 0.33333,-0.33333]	[-0.16667,-0.16667, 0.16667]	[2,7,12,14,19,24]	(1,1)	(2,1)	[0,0,0]
3	[0.33333,-0.33333, 0.33333]	[-0.16667, 0.16667,-0.16667]	[3,8,10,16,18,23]	(1,1)	(3,1)	[0,0,0]
4	[-0.33333, 0.33333, 0.33333]	[0.16667,-0.16667,-0.16667]	[4,6,11,15,20,22]	(1,1)	(4,1)	[0,0,0]

Table 10: 1-th 'H'-'H' [1] (#2) bond cluster (6a@6b), ND, $|\boldsymbol{v}|$= 0.94281 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000,-0.66667,-0.66667]	[0.33333, 0.00000, 0.00000]	[1,-4,17,-20]	(4,1)	(1,1)	[0,0,0]
2	[0.00000, 0.66667,-0.66667]	[-0.33333, 0.00000, 0.00000]	[2,-3,-18,19]	(3,1)	(2,1)	[0,0,0]
3	[-0.66667, 0.00000,-0.66667]	[0.00000, 0.33333, 0.00000]	[5,-8,13,-16]	(3,1)	(1,1)	[0,0,0]
4	[-0.66667, 0.00000, 0.66667]	[0.00000,-0.33333, 0.00000]	[6,-7,-14,15]	(2,1)	(4,1)	[0,0,0]
5	[-0.66667,-0.66667, 0.00000]	[0.00000, 0.00000, 0.33333]	[9,-12,21,-24]	(2,1)	(1,1)	[0,0,0]
6	[0.66667,-0.66667, 0.00000]	[0.00000, 0.00000,-0.33333]	[10,-11,-22,23]	(4,1)	(3,1)	[0,0,0]